

Microsoft — 30 DSA Interview Practice Questions

Targeted practice set for software-engineering interview preparation. Problems are LeetCode-style and can be practiced on LeetCode, GeeksforGeeks, HackerRank, CodeSignal, CodeChef or equivalent platforms. This is a preparation list based on public interview patterns; it is not a claim that these exact questions will be asked.

Public reports describe Arrays/Strings plus Graph/DP and emphasize clean code and edge cases.

#	Problem	Topic	Difficulty	Platform
1	Best Time to Buy and Sell Stock	Array	Easy	LeetCode-style
2	Climbing Stairs	DP	Easy	LeetCode-style
3	Linked List Cycle	Linked List	Easy	LeetCode-style
4	Merge Two Sorted Lists	Linked List	Easy	LeetCode-style
5	Reverse Linked List	Linked List	Easy	LeetCode-style
6	Two Sum	Array/Hashing	Easy	LeetCode-style
7	0/1 Knapsack	DP	Medium	LeetCode-style
8	Add Two Numbers	Linked List	Medium	LeetCode-style
9	Binary Tree Level Order Traversal	Tree/BFS	Medium	LeetCode-style
10	Cheapest Flights Within K Stops	Graph/DP	Medium	LeetCode-style
11	Coin Change	DP	Medium	LeetCode-style
12	Copy List with Random Pointer	Linked List	Medium	LeetCode-style
13	Decode Ways	DP	Medium	LeetCode-style
14	Find Minimum in Rotated Sorted Array	Binary Search	Medium	LeetCode-style
15	House Robber	DP	Medium	LeetCode-style
16	Koko Eating Bananas	Binary Search on Answer	Medium	LeetCode-style
17	Longest Common Subsequence	DP	Medium	LeetCode-style
18	Longest Increasing Subsequence	DP	Medium	LeetCode-style
19	Lowest Common Ancestor of a Binary Tree	Tree	Medium	LeetCode-style
20	Maximal Square	DP/Matrix	Medium	LeetCode-style
21	Maximum Subarray	Array/Kadane	Medium	LeetCode-style
22	Number of Islands	Graph/BFS-DFS	Medium	LeetCode-style
23	Product of Array Except Self	Array	Medium	LeetCode-style
24	Search in Rotated Sorted Array	Binary Search	Medium	LeetCode-style
25	Word Break	DP	Medium	LeetCode-style
26	Binary Tree Maximum Path Sum	Tree/DFS	Hard	LeetCode-style
27	Edit Distance	DP	Hard	LeetCode-style
28	Median of Two Sorted Arrays	Binary Search	Hard	LeetCode-style
29	Merge K Sorted Lists	Heap/Linked List	Hard	LeetCode-style
30	Serialize and Deserialize Binary Tree	Tree/Design	Hard	LeetCode-style

Interview practice rule: For each problem, first explain brute force, then optimize it. Always state time complexity, space complexity, assumptions and edge cases. After solving, practice one variation of the same pattern.